

A Degree-Four Boundary for the v -Number under Integral Closure

Three Variables and at Most Four Minimal Generators

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Abstract

Let K be any field and I an (x, y, z) -primary monomial ideal in $K[x, y, z]$. We prove that

$$v(\bar{I}) \leq v(I)$$

whenever the unique minimal monomial generating set of I has at most four members and every member has total degree at most four. The proof reduces the quantified class, with the variables still labelled, to 517 exponent antichains and exhausts them using exact arithmetic. Primal rational feasibility and dual integer Newton-polyhedron tests agree on every membership decision and on both complete socle lists; a third exact Fourier–Motzkin calculation independently reproduces all v -numbers. The proof is characteristic-free. A known four-generator example of generator degree five has $v(I) = 2 < 3 = v(\bar{I})$, so the degree restriction cannot simply be raised to five.

1 Statement and context

Let $R = K[x, y, z]$ have its standard grading and let $\mathfrak{m} = (x, y, z)$. For a homogeneous ideal $J \subsetneq R$, we use the convention

$$v(J) = \min\{d : \text{there are } f \in R_d \text{ and } \mathfrak{p} \in \text{Ass}_R(R/J) \text{ with } (J : f) = \mathfrak{p}\}.$$

For a monomial ideal with exponent set A , its Newton polyhedron is

$$\text{NP}(I) = \text{conv}(A) + \mathbb{R}_{\geq 0}^3, \quad x^u \in \bar{I} \iff u \in \text{NP}(I) \cap \mathbb{N}^3.$$

These are the conventions used by Biswas, Mandal and Phukan [1]. They prove the comparison inequality in two variables and for three-variable equigenerated monomial ideals, among other classes. Their Example 3.3 gives

$$I = (x^2, y^2, z^5, xyz), \quad v(I) = 2 < 3 = v(\bar{I}).$$

The degree-four question below is a finite boundary suggested by those results; we do not attribute its formulation as an open problem to that paper.

Theorem 1. *Let K be any field and let $I \subset K[x, y, z]$ be a proper $\mathfrak{m}$ -primary monomial ideal. If the unique minimal monomial generating set of I has at most four members, each of total degree at most four, then*

$$v(\bar{I}) \leq v(I).$$

The qualifier “minimal” is essential to the stated boundary: the degree condition is not imposed on an arbitrary redundant presentation. Theorem 1 concerns exactly three variables, primary ideals, and at most four minimal generators; it makes no claim for a larger class.

2 Socles and the complete finite domain

Write $|u| = u_1 + u_2 + u_3$ and let e_1, e_2, e_3 be the coordinate vectors.

Lemma 2 (socle reduction). *If $J \subset R$ is an $\mathfrak{m}$ -primary monomial ideal, then*

$$v(J) = \min\{|u| : x^u \notin J \text{ and } x^{u+e_i} \in J \text{ for } i = 1, 2, 3\}.$$

The same formula applies to $\bar{J}$.

Proof. The only associated prime of R/J is $\mathfrak{m}$. If a homogeneous $f \notin J$ satisfies $\mathfrak{m}f \subset J$, termwise membership in the monomial ideal J shows that every monomial term of f outside J is itself a socle monomial of the same degree. Conversely, if $x^u \notin J$ and all three variable multiples belong to J , then $\mathfrak{m} \subseteq (J : x^u) \subsetneq R$; maximality of $\mathfrak{m}$ gives $(J : x^u) = \mathfrak{m}$. This proves the formula. Integral closure preserves the radical, and the integral closure of a monomial ideal is monomial, so $\bar{J}$ is again $\mathfrak{m}$ -primary and the same argument applies. $\square$

Lemma 3 (parameterization). *Every ideal in Theorem 1, with variable labels retained, is exactly one of*

$$(x^a, y^b, z^c), \quad (x^a, y^b, z^c, x^r y^s z^t), \quad (1)$$

where $1 \leq a, b, c \leq 4$, and in the second case

$$0 \leq r < a, \quad 0 \leq s < b, \quad 0 \leq t < c, \quad r + s + t \leq 4, \quad |\text{supp}(r, s, t)| \geq 2. \quad (2)$$

There are 64 ideals of the first form and 453 of the second form, hence 517 in all.

Proof. Primaryness implies that a power of each variable belongs to I . A minimal monomial generator dividing a pure power is itself a pure power, and minimality permits exactly one such generator on each coordinate axis. At most one further generator remains. It is incomparable with all three pure powers exactly when the three strict inequalities and the support condition in (2) hold. The degree hypotheses give the remaining bounds. This proves both necessity and sufficiency.

There are $4^3 = 64$ choices without a mixed generator. For a mixed exponent of support two, choose its missing coordinate in three ways. If its positive coordinates are p, q , then $p, q \geq 1$, $p + q \leq 4$, and the number of axis-power choices is $4(4 - p)(4 - q)$. Since

$$\sum_{p, q \geq 1, p+q \leq 4} (4 - p)(4 - q) = 31,$$

this gives $3 \cdot 4 \cdot 31 = 372$ ideals. Full support allows only $(1, 1, 1)$, contributing $3^3 = 27$, and the three permutations of $(2, 1, 1)$, contributing $3(2 \cdot 3 \cdot 3) = 54$. Thus the mixed count is $372 + 27 + 54 = 453$. No variable-permutation quotient has been used. $\square$

For (1), set

$$B(a, b, c) = \{0, \dots, a - 1\} \times \{0, \dots, b - 1\} \times \{0, \dots, c - 1\}.$$

Every exponent outside this box already dominates a pure-power generator and therefore belongs to $I \subseteq \bar{I}$. Hence all standard monomials and all socle exponents of either ideal occur in $B(a, b, c)$. Testing the expanded grid $B^+ = \{0, \dots, a\} \times \{0, \dots, b\} \times \{0, \dots, c\}$ also includes every immediate successor needed in Lemma 2.

3 Two exact exhaustive certificates

Let $A = \{\alpha_1, \dots, \alpha_m\} \subset \mathbb{N}^3$ be the generator exponents, where $m \in \{3, 4\}$. The first certificate uses the exact equivalence

$$u \in \text{NP}(I) \iff \begin{cases} \lambda_i \geq 0, & \sum_i \lambda_i = 1, \\ \sum_i \lambda_i \alpha_i \leq u & \text{coordinatewise} \end{cases} \quad \text{for some } \lambda. \quad (1)$$

For fixed u , the feasible set in the simplex is a compact rational polytope. If it is nonempty, it has a vertex. Within the affine hyperplane $\sum_i \lambda_i = 1$, some $m - 1$ linearly independent active inequalities among the m nonnegativity inequalities and three coordinate inequalities determine that vertex. The program enumerates every such subset, solves the square system by exact rational Gaussian elimination, and checks all original inequalities. Thus it decides (1) without a numerical tolerance. It decides membership in I by exponent divisibility and computes both socles by the three-successor test of Lemma 2.

The second certificate starts from the exact dual characterization

$$u \in \text{NP}(I) \iff w \cdot u \geq \min_{\alpha \in A} w \cdot \alpha \quad \text{for every } w \in \mathbb{R}_{\geq 0}^3. \quad (2)$$

Necessity is immediate, and sufficiency follows from separation because the recession cone of $\text{NP}(I)$ is $\mathbb{R}_{\geq 0}^3$. The coordinate planes and the generator-tie planes $w \cdot (\alpha_i - \alpha_j) = 0$ subdivide the nonnegative orthant into polyhedral cones on each of which the minimum in (2) is one fixed linear form. Each extreme ray is cut out by at least two boundary planes. The program therefore forms all pairwise integer cross products of their normals, retains the nonnegative rays, and checks (2) on them. Every point of a subdivision cone is a nonnegative combination of its extreme rays, so these integer tests are necessary and sufficient. Extra retained nonnegative rays are harmless because (2) is required for every nonnegative w .

This second route also computes the two socles differently. Without a mixed generator the input socle is $\{(a - 1, b - 1, c - 1)\}$. With mixed exponent $q = (r, s, t)$, its elements are, for each $q_i > 0$, the point whose i -th coordinate is $q_i - 1$ and whose other coordinates are their box maxima. Indeed, every standard exponent fails to dominate q in some such coordinate and is dominated by the corresponding listed point; the listed points are maximal. For $\bar{I}$, the program constructs the complete standard down-set in B and selects its globally maximal points. Global maximality in a finite down-set is equivalent to the three-successor socle condition.

Proposition 4 (exact execution record). *The two implementations enumerate the same 517 labelled ideals. Each checks 31,922 expanded-grid exponents, containing all 13,270 standard-box exponents, and they agree on all 63,844 input/closure membership bits, on every complete socle list, and on every pair $(v(I), v(\bar{I}))$. Their common distribution is*

$v(\bar{I}) - v(I)$	−6	−5	−4	−3	−2	−1	0
number of ideals	1	6	19	82	194	124	91

In particular, no difference is positive.

The comparison is record-by-record, rather than a comparison of aggregate counts. A third implementation substitutes one barycentric variable in (1) and applies exact rational Fourier–Motzkin elimination. It recomputes both socles by successor tests and reproduces all 1,034 complete socle lists (3,552 listed exponents) and all 517 pairs of v -numbers. This route uses neither active-set vertex

enumeration nor dual-ray construction. All three implementations also return $(v(I), v(\bar{I})) = (2, 3)$ for (x^2, y^2, z^5, xyz) , the degree-five control from [1].

For methodological comparison only, a similar proof-engineering pattern—a structural reduction followed by canonical finite enumeration, exact arithmetic and an independent verifier—appears in Zeng’s work on a different problem about characteristic-zero monomial algebras [2]. That paper contains no claim about integral closure, v -numbers, or the boundary proved here.

Proof of Theorem 1. Lemma 3 places every quantified ideal in exactly one of the 517 labelled records. The box argument contains every possible socle exponent. Equations (1) and (2), together with the correctness arguments above, decide every integral-closure membership exactly, while Lemma 2 converts the complete socles into the two v -numbers. Proposition 4 has no positive difference, so $v(\bar{I}) \leq v(I)$ for every record and hence for every ideal in the theorem. All inputs to the calculation are integer exponent comparisons and exact rational operations. They do not depend on K , so the conclusion holds over every field and in every characteristic. $\square$

4 Reproducibility and limitations

The supplementary certificate consists of two enumerators and a comparison script. From the project directory, the decisive records are regenerated by

```
python3 evidence/enumerate_lp.py evidence/lp_results.json
python3 evidence/enumerate_facets.py evidence/facet_results.json
python3 evidence/compare_runs.py evidence/lp_results.json \
    evidence/facet_results.json evidence/verification.json
```

The first implementation uses Python’s exact `fractions.Fraction`; the second uses integers only. Neither uses floating point. The generated JSON records the generator exponents, both membership bitmaps on B^+ , both full socle lists, and both v -numbers for every labelled ideal. The independent Fourier–Motzkin checker is included as `audit/referee_independent.py`.

The computation proves only the quantified finite theorem. It does not classify the 91 equality cases, address five or more minimal generators, or assert the inequality for arbitrary three-variable ideals. The cited degree-five example shows that replacing the degree bound four by five would make the stated universal assertion false.

Computational and AI assistance. Exact Python programs were used for the exhaustive membership and socle calculations; no formal proof assistant was used. An AI-assisted workflow helped formulate the finite reduction, implement and cross-check the programs, audit the mathematical scope, and prepare the manuscript. The theorem rests on the explicit reduction, the exact algorithms justified above, and the reproducible record-level certificates, not on numerical fitting or an AI confidence assessment.

References

- [1] Prativa Biswas, Mousumi Mandal, and Partha Phukan. A comparison of the v -number of a monomial ideal and its integral closure, 2026. arXiv:2609.05044v1.
- [2] Zijian Zeng. The whiskered-graph weak lefschetz conjecture for graphs on eight vertices. SSRN preprint, Elsevier BV, 2026. DOI: 10.2139/ssrn.7385138.